

MARYAM BANO

Full-Stack Web Developer • AI & Machine Learning Enthusiast • Freelancer
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◆ PROFESSIONAL SUMMARY

Results-driven Full-Stack Web Developer and Computer Science undergraduate with hands-on experience delivering 10+ end-to-end applications spanning web development, data analytics, and applied machine learning. Built production systems for Bonanza Satrangi including an AI-powered retail sales chatbot, an internal voucher reporting platform, and an employee complaint management system using Python, Flask, and Oracle Database. Completed a Data Analysis internship at Excelerate, developing a Random Forest Classifier that achieved 100% accuracy on a held-out test set of 1,700+ records. Freelance developer with a track record of shipping complete e-commerce platforms with cart/checkout flows, automated PDF invoicing, and email notifications. Certified in Generative AI with a 97.25% Top Performer score. Seeking to bring strong full-stack and AI/ML skills to a data-driven engineering team.

◆ EDUCATION

Bahria University, Karachi Campus, Pakistan

Sep 2023 – Dec 2027

Bachelor of Computer Science

Bahria College, Karachi, Pakistan

Sep 2021 – Jul 2023

Intermediate in Computer Science | Position Holder

◆ WORK EXPERIENCE

IT Intern — Bonanza Satrangi

Jul 2026 – Aug 2026 (2 Months)

Retail Sales AI Chatbot

- Designed and built an AI-powered Streamlit dashboard that lets business stakeholders ask plain-English questions about retail sales data and receive instant answers on top stores, best-selling categories, and daily sales trends.
- Integrated the Google Gemini API with a Pandas-based query engine to translate natural-language questions into structured data queries, cutting the time needed to generate ad-hoc sales insights.

Tech Stack: Python, Streamlit, Pandas, Google Gemini API, NLP, Data Visualization

Employee Complaint Management System (ECMS)

- Implemented role-based authentication for Employee and IT/Admin users, complaint submission with file attachments, and status/search-based complaint tracking.
- Built employee and IT/Admin dashboards with complaint statistics, plus date-based reporting with export to PDF and Excel.
- Integrated Oracle Database for secure, centralized data management across the application.

Tech Stack: Python, Flask, Oracle Database, HTML, CSS, Jinja2, oracledb, openpyxl, ReportLab

Gift Voucher Report Generation System

- Built an internal voucher management system enabling bulk voucher upload via Excel, with validation and automated insertion into an Oracle Database.
- Designed batch-based and voucher-code-range report generation, allowing staff to produce targeted PDF voucher batches on demand.
- Integrated JasperReports with the Flask backend via a Java-based report execution layer to generate multi-page branded PDF vouchers with QR/barcode components.

Tech Stack: Python, Flask, Oracle Database, JasperReports, Java 21, Pandas, openpyxl

AI Powered Data Analysis Intern (Remote) — Excelerate

Jun 2026 – Jul 2026 (1 Month)

- Built and evaluated a Random Forest Classifier to predict student engagement using a dataset of 8,558 records and 24 features, achieving 100% accuracy on a held-out test set of 1,712 records.
- Executed the full data science pipeline — data cleaning, preprocessing, and exploratory data analysis (EDA) — to uncover participation trends across demographic and opportunity-related features.

- Ran feature importance analysis and generated probability-based predictions to identify the strongest drivers of student engagement, visualized via confusion matrix and probability heatmap.
- Authored an executive-ready Data Understanding Report with actionable, KPI-backed recommendations for improving participation, presented to program stakeholders at the end of the internship.

Tech Stack: Python, Pandas, NumPy, Scikit-learn, Random Forest, Matplotlib, Seaborn, EDA, Feature Engineering

◆ TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, TypeScript, Java, HTML, CSS, Tailwind CSS, R
- **Web & Backend:** Flask, Django, Next.js, SQLAlchemy, Jinja2, REST APIs, Resend API, Streamlit
- **AI & Machine Learning:** NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Random Forest, Google Gemini API, NLP
- **Databases:** Oracle Database, MySQL, SQLite, Microsoft SQL Server (SSMS), MS Access
- **Reporting & Documents:** ReportLab, JasperReports, PDF Generation, Excel Automation (openpyxl)
- **Tools & Platforms:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Anaconda
- **Other:** Problem Solving, Team Collaboration, Research & Presentation, MS Excel, Word, PowerPoint

◆ PROJECTS

Enterprise Payroll Hub | [GitHub ↗](#)

- Built a production-ready enterprise payroll management web application with automated payslip generation, attendance tracking, and configurable corporate salary structures.

Tech Stack: Next.js, TypeScript, Tailwind CSS

Mariame Plants Ecosystem (Freelance Project) | [GitHub ↗](#)

- Built a full-stack e-commerce web application with a product catalog, category filters, and shopping cart functionality.
- Developed secure checkout with SQLite, SQLAlchemy, and automated PDF invoice generation.
- Integrated Resend API for email notifications and deployed a responsive UI on Vercel.

Tech Stack: Python, Flask, SQLAlchemy, SQLite, ReportLab, Resend API, HTML5, CSS3

E-Commerce Product Analytics & ML Dashboard | [GitHub ↗](#)

- Built an end-to-end analytics pipeline scraping e-commerce data via Selenium, cleaning and performing EDA with Pandas and NumPy, applying PCA, and training ML models with Scikit-learn.
- Deployed an interactive Streamlit dashboard with Plotly visualizations for exploring product and pricing trends.

Tech Stack: Python, Selenium, Scikit-learn, Streamlit

◆ CERTIFICATIONS & TRAINING

- **DataCamp Full Scholarship** — Bahria University Developer Society (1 Year): Full one-year access to DataCamp's Data Science and Analytics platform; completed structured career tracks in data analysis, Python development, and machine learning.
- **NCEAC-HEC Generative AI Training, Cohort 3 (Mar–May 2026), Pak Angels:** Advanced training in Generative AI, Advanced Python, and AI application development; Top Performer Certificate with a final score of 97.25%.
- **Governor Sindh Initiative for GenAI, Web3 & Metaverse (2023–Present):** Agentic AI, Advanced Python, TypeScript, Next.js, and Tailwind CSS.
- **Competition Certificates:** Code in Dark and Speed Programming competitions, Bahria University.

◆ AWARDS & ACTIVITIES

- Position Holder — Computer Science Department, Bahria College (2022).
- Delegate — Model United Nations (2021).

◆ LANGUAGES

English (Fluent) | Urdu (Native)